Approaches to time series

There are two approaches to the decomposition of time series data

(i) Additive approach

(ii) Multiplicative approach

The above two approaches are used in decomposition, depending on the nature of relationship among the four components.

The additive approach

The additive approach is used when the four components of a time series are visualized as independent of one another. Independence implies that the magnitude and pattern of movement of the components do not affect one another. Under this assumption the magnitudes of the time series are regarded as the sum of separate influences of its four components.

$$Y = T + C + S + R$$

where Y = magnitude of a time series

T = Trend,
C =Cyclical component,
S =Seasonal component, and
R = Random component

In additive approach, the unit of measurements remains the same for all the four components.

The Multiplicative approach

The multiplicative approach is used where the forces giving rise to the four types of variations are visualized as interdependent. Under this assumption, the magnitude of the time series is the product of its four components.

i.e. $Y = T \times C \times S \times R$

Difference between the two approaches

Multiplicative	Additive
(i) Four components of time series are interdependent	Four components of time series are independent
(ii) Logarithm of components are additive	Components are additive

7.3 MEASEUREMENTS OF COMPONENTS

(i) Secular trend

It refers to the long term tendency of the data to move in an upward or downward direction. For example, changes in productivity, increase in the rate of capital formation, growth of population, *etc.*, follow secular trend which has upward direction, while deaths due to improved medical facilities and sanitations show downward trend. All these forces occur in slow process and influence the time series variable in a gradual manner.

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